

Tutorix: An AI Toolkit for Bangladeshi Teachers and Students, Grounded in Their Own Textbooks

Scientific field	Robotics & AI
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Supervisor	Md. Ashrafuzzaman
Type	Working web application (teacher, student and admin roles; English, Bangla and bilingual)

Project Abstract

Teachers in Bangladesh spend many hours each week preparing lesson plans, question papers, answer keys, rubrics and report card comments, while many students, especially outside big cities, have no tutor to explain difficult topics or guide their revision. Tutorix is a web platform that addresses both problems with 20 AI tools, 10 for teachers and 10 for students. Teachers can create lesson plans, Bloom's taxonomy-aligned worksheets, marking schemes, differentiated content and term plans in seconds, and can check test quality with classical item analysis. Students get smart notes, flashcards, practice quizzes with weak-topic diagnosis, mind maps, step-by-step problem solving, concept explanations, mnemonics, study planners and an English-Bangla vocabulary builder. Every tool works in English, Bangla or both. General AI chatbots often invent facts, so Tutorix grounds its answers in the user's own textbook or notes. Short material is read in full through Cache-Augmented Generation, and long material, even a whole book, is searched through Retrieval-Augmented Generation with hybrid keyword and semantic retrieval, so only the relevant passages are used. The system runs on Supabase (PostgreSQL with pgvector, authentication and edge functions) and Groq-hosted open-source language models. All results are saved to a personal library and can be exported as clean PDF or Word documents.

Objectives

The main objective of Tutorix is to cut teachers' preparation time by automating lesson plans, question papers, answer keys, rubrics and feedback comments while keeping them aligned with the curriculum. It also aims to give every student a personal AI tutor that explains, quizzes and plans revision in English or Bangla. The project seeks to keep AI answers accurate and syllabus-relevant by basing them on the user's own textbook or notes through RAG and CAG rather than general internet knowledge. It promotes higher-order thinking by mapping questions to Bloom's taxonomy, and helps teachers improve their assessments with item analysis measures such as difficulty, discrimination, KR-20 reliability and distractor performance. Finally, it aims to stay affordable and fast enough for low-bandwidth, low-budget schools through open-source models and response

caching, while protecting user data with role-based access and row-level security so that each user sees only their own work.

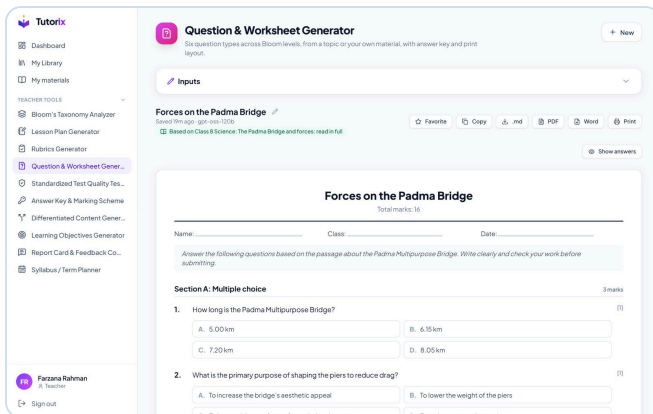
Innovation / Novelty

The main novelty of Tutorix is that it automatically picks the right way to ground its answers in the user's material. Short material is read in full through Cache-Augmented Generation. Long material goes through Retrieval-Augmented Generation, which combines BM25 keyword search with semantic vector search using reciprocal rank fusion. Most embedding models handle Bangla poorly, so Tutorix matches Bangla text with a Unicode-aware keyword search, which lets Bangla textbooks work as well as English ones. Unlike existing single-purpose apps, it serves both teachers and students on one platform with a shared library and export system. Beyond AI review, its test quality tool calculates real psychometric statistics from student responses, which is rare among AI education tools. It is built for the local context, with Bangladeshi class levels from Class 1 to HSC and O/A Level, bilingual output and Bangladeshi examples. It also stays reliable and low-cost through automatic model and API-key fallback, sanitised output without broken formatting or math, and a seven-day response cache that answers repeated requests instantly at no extra cost.

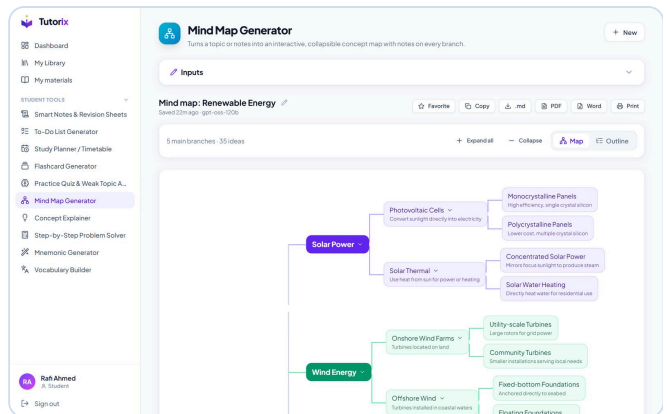
Expected Impact

Tutorix is expected to reduce teachers' preparation work from hours to minutes, giving them more time for teaching and individual student support, while making their assessments more balanced across Bloom's levels and more reliable. Students will get free, round-the-clock tutoring in their own language based on their own textbook, which especially helps rural and low-income students who cannot afford private coaching and so reduces the learning gap. Schools will benefit from better question quality and data-driven insight into weak topics, supporting the competency-based approach of Bangladesh's new national curriculum. Because it runs in any web browser on low-cost infrastructure, it can scale to thousands of schools and later expand to more subjects, local exam boards and offline or mobile use. It also has research value, showing how grounding methods like RAG and CAG can make AI trustworthy for education in low-resource languages such as Bangla.

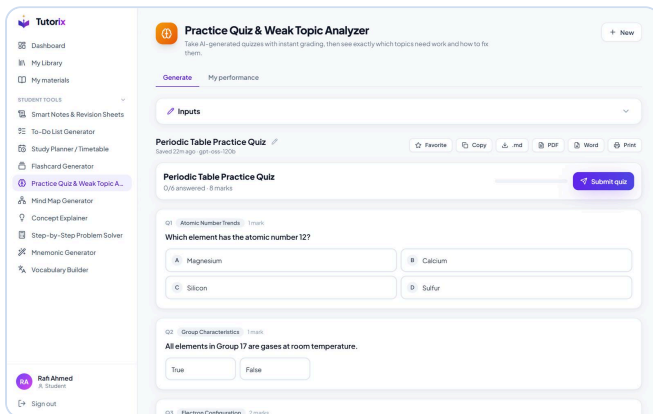
Tutorix at a glance



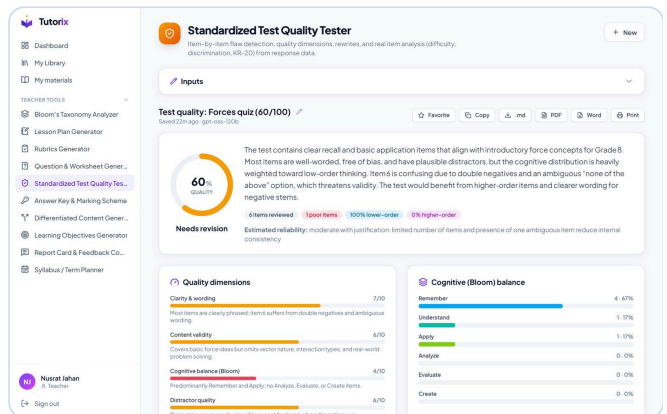
Teacher worksheet from a saved Padma Bridge chapter
("Based on" badge)



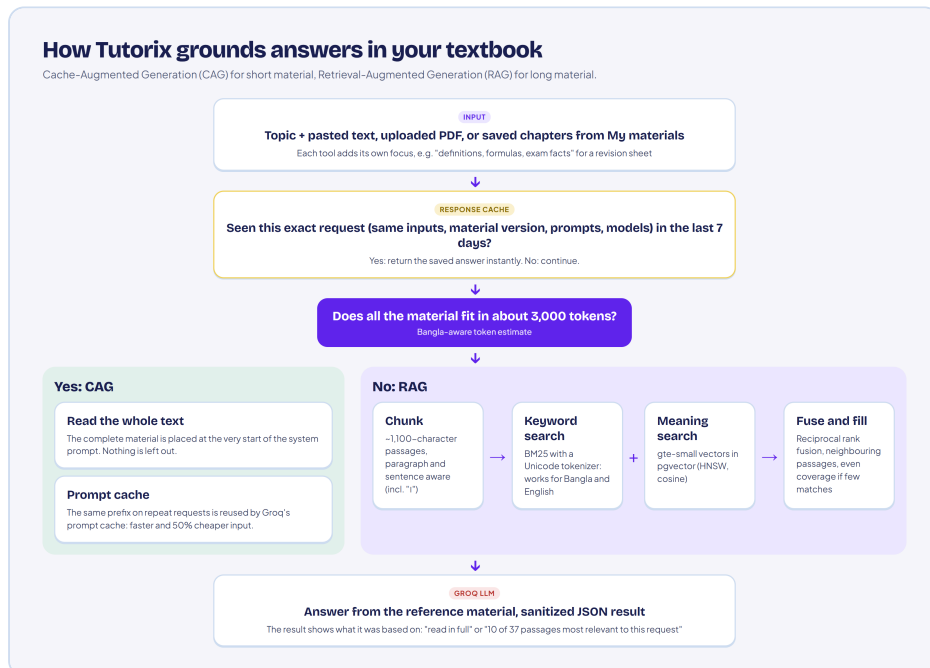
Student mind map, interactive and collapsible



Practice quiz with instant grading and weak-topic analysis



Standardized test quality tester with item analysis



How Tutorix grounds answers in the user's own material (CAG and RAG)